

1 Limits and continuity

Definition 1. A function $f: \mathbb{R} \rightarrow \mathbb{R}$ is *continuous at a* if for every $\varepsilon > 0$ there exists $\delta > 0$ such that $|x - a| < \delta$ implies $|f(x) - f(a)| < \varepsilon$.

Lemma 2. If f and g are continuous at a , then so is $f + g$.

Proof. Let $\varepsilon > 0$. Choose $\delta_1, \delta_2 > 0$ so that $|x - a| < \delta_1$ implies $|f(x) - f(a)| < \varepsilon/2$ and $|x - a| < \delta_2$ implies $|g(x) - g(a)| < \varepsilon/2$. Put $\delta = \min\{\delta_1, \delta_2\}$. Then for $|x - a| < \delta$,

$$\begin{aligned} |(f + g)(x) - (f + g)(a)| &\leq |f(x) - f(a)| + |g(x) - g(a)| \\ &< \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon. \end{aligned} \quad \square$$

Theorem 3. The function $f(x) = x^2$ is continuous at every $a \in \mathbb{R}$.

Proof. Fix a and $\varepsilon > 0$, and let

$$\delta = \min \left\{ 1, \frac{\varepsilon}{2|a| + 1} \right\}.$$

If $|x - a| < \delta$ then $|x| < |a| + 1$, so $|x + a| < 2|a| + 1$ and

$$|x^2 - a^2| = |x - a| |x + a| < \delta (2|a| + 1) \leq \varepsilon. \quad \square$$

Theorem 4 (Intermediate value theorem). Let $f: [a, b] \rightarrow \mathbb{R}$ be continuous with $f(a) < 0 < f(b)$. Then $f(c) = 0$ for some $c \in (a, b)$.

Proof. Let $S = \{x \in [a, b] : f(x) < 0\}$ and $c = \sup S$, which exists because S is nonempty and bounded above by b . We rule out the two alternatives to $f(c) = 0$.

Suppose $f(c) < 0$. By continuity there is $\delta > 0$ with $f(x) < 0$ on $(c - \delta, c + \delta) \cap [a, b]$, and $c < b$ since $f(b) > 0$; so some point greater than c lies in S , which is impossible.

Suppose $f(c) > 0$. Then $f > 0$ on some interval $(c - \delta, c]$, so $c - \delta$ is an upper bound of S smaller than c , which is again impossible. Hence $f(c) = 0$, and $c \neq a, b$. $\square$

Theorem 5. Let $f(x) = \begin{cases} x \sin(1/x) & x \neq 0, \\ 0 & x = 0. \end{cases}$ Then f is continuous at 0.

Proof. For $x \neq 0$ we have $|f(x) - f(0)| = |x| |\sin(1/x)| \leq |x|$, so $\delta = \varepsilon$ works. $\square$